

Technology Stack

Date: 30 September 2026

Team ID: SWTID-2026-4040

Project Name: ComicCraft — AI Comic Story Creator Using Gemini Models

Team Leader: Bala Esakki R

Team Members: Sandhiya D, Rithikaa B, Anitha R, Anbuselvi

Team Size: 5

Maximum Marks: 2

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Presentation	HTML + Jinja2 templates	Provides browser forms and preview pages with lightweight server-rendered UI.
2	Backend / API	Python + FastAPI	Asynchronous web framework used by the uploaded application and route layer.
3	AI Story Generation	Google Generative AI; Gemini 1.5 Flash / Pro	Flash creates the outline; Pro expands it into narration and dialogue.
4	Image Generation	Diffusers + Stable Diffusion v1-5 + PyTorch	Generates panel artwork from image prompts.
5	Templating	Jinja2	Renders index, preview and export-success HTML pages.
6	File / PDF Support	FPDF, Pillow	Included in requirements for export/image handling.
7	Server	Uvicorn	ASGI server suitable for running FastAPI.
8	Version Control	Git / GitHub	Repository-based source management.